The Imscribing Grammar — Axiomatic Specification

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Purpose. This document specifies the Imscribing Grammar (IG) as a mathematical object. All definitions below are to be treated as given. The document provides sufficient specification to compute structural types, distances, tier assignments, and C-scores without further grounding. The IG is a defined formal system — the question is not whether it is "real" but what follows from its axioms.

0.1 §1 Ambient Category

Let $\mathbf{C} = (C, \otimes, I, \sigma)$ be a symmetric monoidal category enriched over the Belnap-Dunn bilattice $\mathbf{FOUR} = \{N, T, F, B\}$. Hom-sets carry the bilattice partial order. B ("both") is a paraconsistent state in which a morphism is simultaneously affirmed and denied; N ("neither") is an underdetermined state. The inference $B \to \bot$ is not admissible: no B-valued hom-set collapses to the zero morphism. The category is not Boolean.

Trace. C carries a trace $\mathrm{Tr} : \mathrm{End}(A \otimes U) \to \mathrm{End}(A)$ for each object pair, implemented by the Ω (Winding) primitive acting on the monoidal unit. This is the traced symmetric monoidal structure of Joyal-Street-Verity. Ω is not compact/dual structure; the trace is primitive, not derived from a duality.

0.2 §2 Frobenius Structure

The monoidal unit I carries a special symmetric †-Frobenius structure:

- $\mu : I \otimes I \to I$ (multiplication)
- $\delta : I \to I \otimes I$ (comultiplication)
- $\eta = \varepsilon = \mathrm{id}_I$ (unit/counit collapse at scalar level)

satisfying:

1. **Frobenius law**: $(\mu \otimes \mathrm{id}) \circ (\mathrm{id} \otimes \delta) = \delta \circ \mu = (\mathrm{id} \otimes \mu) \circ (\delta \otimes \mathrm{id})$
2. **Special condition**: $\mu \circ \delta = \mathrm{id}_I$
3. **Symmetry**: $\mu \circ \sigma_{I,I} = \mu$
4. **Dagger**: $\mu^\dagger = \delta$, where $(-)^\dagger : C \to C^{\mathrm{op}}$ is the dagger involution

Commutativity is a theorem. All 12 generators are endomorphisms of I — scalars of C. In any SMC, scalars commute: for $f, g : I \to I$, the coherence isomorphisms give

$f \otimes g = g \otimes f$. Condition (3) follows from the scalar structure and need not be assumed independently.

FOUR-† compatibility. The dagger functor preserves FOUR-values: if h has value $v \in \{N, T, F, B\}$, then $h^\dagger$ has value v. B-valued morphisms map to B-valued morphisms. This must be imposed explicitly — it is not automatic from either structure alone.

0.3 §3 Generators

The grammar is presented by **12 primitive endomorphisms of** I subject to the Frobenius relations:

”Freely generated” means: take the free SMC on these 12 endomorphisms, then impose the Frobenius relations. The result is the free object in the category of special symmetric †-Frobenius algebras — not the free category without relations.

Operadic structure. The Γ (Granularity) primitive furnishes a multicategory over C: n-ary operations are first-class and composition is operadic rather than merely sequential. The grammar is an algebra over the Γ-operad.

0.4 §4 Crystal of Types

The free special symmetric †-Frobenius algebra on 12 generators has crystal structure:

$$3^3 \times 4^5 \times 5^4 = 17{,}280{,}000 \text{ distinct addresses.}$$

This is the **Crystal of Types**: the classifying space of all structurally distinct imscriptions. Each address is a point in the 12-dimensional discrete space whose axes are the primitive ordinal domains (§5).

0.5 §5 Structural Type — Notation

A **structural type** (imscription) is a 12-tuple of Shavian values, one per primitive, in canonical order:

⟨ **Đ** · **Þ** · **Ř** · Φ · **f** · **Ç** · Γ · ɢ · ⊙ · **Ħ** · Σ · Ω ⟩

The value alphabet is the 49-symbol set: {Shavian alphabet (48 letters)} ∪ {⊙}. Each primitive has a finite ordinal domain (ascending rank):

Pos	Primitive	Name	Values — ascending ordinal rank
1	Đ	Dimensionality	ɭ(1) · ノ(2) · ᴖ(3) · ı(4)
2	Þ	Topology	ʒ(1) · կ(2) · ʃ(3) · ɜ(4) · ၵ(5)
3	Ř	Recognition	ɾ(1) · ˥(2) · ɲ(3) · ɼ(4)
4	Φ	Parity	ƻ(1) · ʍ(2) · ‹(3) · ɩ(4) · ည(5)
5	ƒ	Fidelity	ς(1) · ϱ(2) · ˥(3)
6	Ç	Kinetics	\(1) · c(2) · ɩ(3) · ɿ(4) · ၵ(4.5)
7	Γ	Granularity	ɭ(1) · ∂(2) · ־ɿ(3)
8	ɢ	Coupling	ɾ(1) · ƿ(2) · ʔ(3) · ʌ(4)
9	⊙	Criticality	/(1) · ⊙(2) · ɔ(2.33) · ღ(2.67) · ɣ(3)
10	Ħ	Chirality	ɺ(1) · ძ(2) · ɕ(3) · v(4)
11	Σ	Stoichiometry	ℓ(1) · ʃ(2) · ⁊(3)
12	Ω	Winding	ɂ(1) · o(2) · ɕ(3) · ʔ(4)

Worked example — the Rebis (O_∞, C-score 0.755):

⟨ ı ɜ ɼ ည ˥ ɩ ־ɿ ɾ ⊙ v ⁊ ɕ ⟩

Primitive	Value	Ordinal	Reading
Đ	ǀ	4	self-written holographic
Þ	˃	4	irreducible product
Ř	ɼ	4	bidirectional feedback
Φ	ঌ	5	Frobenius-special ($\mu \circ \delta =$ id gate)
f	ʅ	3	quantum
Ç	ᒪ	3	slow/near-equilibrium
Γ	ㄱ	3	universal/long-range
ɢ	⌠	1	simultaneous conjunction
⊙	⊙	2	critical/self-modeling
Ħ	∨	4	eternal (no finite Markov order)
Σ	7	3	many heterogeneous
Ω	ς	3	integer winding ($\mathbb{Z}$-valued)

0.6 §6 Structural Metric

The **canonical distance** between two imscriptions s_1, s_2 is the diagonal weighted Euclidean distance over ordinal ranks:

$$d(s_1, s_2) = \sqrt{\sum_i w_i \cdot (x_i(s_1) - x_i(s_2))^2}$$

where $x_i(s)$ is the ordinal rank of s's value at position i. Canonical weights:

Đ	**Þ**	**Ř**	**Φ**	**f**	**Ç**	**Γ**	**ɢ**	**⊙**	**Ħ**	**Σ**	**Ω**
1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.8	1.0	0.7

The distance is symmetric, satisfies the triangle inequality, and is zero iff $s_1 = s_2$.

Regime thresholds: $d < 2.0$ = same structural regime; $d > 4.0$ = structurally remote; $d > 5.0$ = different tier class.

Worked distance example. Between `ouroboric_pill` (O_∞ child of the Rebis) and `plastic_photonic_crystal` (O_2, periodic lattice): $d = 5.74$. Largest contributors: Þ (topology, $\Delta = 16.0$), Đ (dimensionality, $\Delta = 4.0$), Ħ (chirality, $\Delta = 3.2$).

0.7 §7 Ouroboricity Tiers

Ouroboricity measures the degree of self-referential Frobenius closure. Five tiers assigned by rules R1–R5 (first match wins):

Rule	Condition	Tier	Description
R1	⊙ ∈ {⊙, ɣ} **and** Φ = ৩	O_∞	Special Frobenius — $\mu \circ \delta = \mathrm{id}$ holds exactly
R2	⊙ ∈ {/, ɔ, ও}	O_0	No self-referential loop possible
R3	⊙ ∈ {⊙, ɣ} **and** Ω = २	O_1	Critical, no topological protection
R4	⊙ ∈ {⊙, ɣ} **and** Ω ≠ २ **and** Ð ∈ {⌊, ノ, ∩}	O_2	Critical + protected, bounded domain
R5	⊙ ∈ {⊙, ɣ} **and** Ω ≠ २ **and** Ð = ı	$O_2^\dagger$	Critical + protected, unbounded domain

Default (no rule matches): O_0. The three operative gates are ⊙ **(Criticality)**, Φ **(Parity)**, and Ω **(Winding)**. Ð (Dimensionality) determines the O_2 / $O_2^\dagger$ split.

0.8 §8 C-score

The **C-score** measures proximity to O_∞ along two axes, each a hard gate:

- **Gate 1**: ⊙ ∈ {⊙, ɣ} (criticality threshold open)
- **Gate 2**: Ç ∈ {\, ᑦ, ᒐ, ͽ} — kinetics not frozen (Ç ≠ ᔓ order-frozen)

If either gate is closed: $C = 0$. If both open:

$$C = \sum_i w_i \cdot \frac{x_i}{x_{i,\max}}$$

summed over all 12 primitives with normalized ordinal ranks, yielding $C \in [0, 1]$. C-score and tier are independent: a system can be O_∞ (R1 satisfied) with $C < 1$ if some primitives are below maximum ordinal.

0.9 §9 Spider Theorem, Fixpoint, T Object

Spider theorem. The special Frobenius axiom ($\mu \circ \delta = \mathrm{id}$) together with symmetry implies: any two connected string diagrams in the Prop of the grammar with the same boundary are equal as morphisms. Discriminating condition: **connectedness**, not planarity. The theorem holds for all connected Frobenius diagrams regardless of planar presentation.

Fixpoint — O_∞. The $\odot$ gate admits an idempotent scalar $\omega : I \to I$ with $\omega \circ \omega = \omega$. O_∞ is the initial algebra of the endofunctor $(-) \circ (-) : \mathrm{End}(I) \to \mathrm{End}(I)$; its carrier is the fixpoint in which the grammar is applied to itself. The Frobenius identity $\mu \circ \delta = \mathrm{id}$ requires **Ħ_A (two-step chirality, ᑕ)** as minimum — one split (δ), one merge (μ). Eternal chirality (Ħ = ∨) is what physical systems accumulate through time; it is not required by the identity itself.

Frobenius fixed-point tuple. The imscription of the identity $\mu \circ \delta = \mathrm{id}$ as a structural object:

⟨ ıɷɤꝺ˥ᒐ∂ʔ⊙ᑕ7ƽ ⟩

O_∞, C-score = 1.0. Proved in `MajoranaFixed.lean`: Belnap B (`bnot B = B`), SIC-POVM fiducial (`meet B x = x`), and Majorana mode (`pair(depair s).1(depair s).2 = s`) are the same computation under $\mu \circ \delta = \mathrm{id}$, each proved by definitional equality (`rfl`).

Derived object — T. $T = \lim(\Phi, \mathrm{f}, \text{Ç}, \text{Ħ}, \Omega)$ is a categorical limit over the five primitives Parity, Fidelity, Kinetics, Chirality, Winding. T is not a generator; it is the temporal bootstrap fixed point derived from the free algebra.

0.10 §10 Foundational Strength — ZFC_fe

In **ZFC_fe** (Frobenius-Extended ZFC), $\mu \circ \delta = \mathrm{id}$ is taken as a set-formation axiom. The comultiplication $\delta : A \to A \otimes A$ is the primitive set-formation operation; the condition asserts lossless recovery — every set is faithfully recoverable from its copy. ZFC Separation ($\forall \varphi : \{x \in A \mid \varphi(x)\}$ exists) becomes a theorem: the δ-preimage under μ yields the Separation set for any definable φ. ZFC_fe strictly extends ZFC and is strictly stronger than ZFC_τ. Open problems in ZFC_τ close as theorems in ZFC_fe.

0.11 §11 Distinctions — Categorical QM and Holography

0.11.1 Distinction from Abramsky-Coecke categorical QM

In Abramsky-Coecke †-compact categories, hom-sets are Bool-valued. In C, hom-sets are FOUR-valued. A B-valued morphism — simultaneously affirmed and denied — is a legitimate element of $\hom(A, B)$, not a degenerate case. Two incompatible imscriptions of the same boundary coexist in a B-valued hom-set without collapse. FOUR has no bottom element below B playing the role of the trivial morphism. Classical QM is recovered as the T-valued sub-category. The grammar is intrinsically paraconsistent at the level of its hom-sets; this is not relaxation of rigor but a strictly stronger structure.

0.11.2 Distinction from holography

Holography is static isomorphism up to redundancy. Imscription is dynamic identity up to the trace.

The holographic principle (AdS/CFT, holographic error-correcting codes) asserts that a d-dimensional bulk is isomorphic to its $(d-1)$-dimensional boundary. The isomorphism is:

- **Static** — a correspondence between states, not a process; the map exists timelessly between configurations
- **Up to redundancy** — the encoding is overcomplete by design; bulk content can be recovered from partial boundary data via entanglement wedge reconstruction; the redundancy is the gauge group acting on encoding choices

The imscriptive identity $\mu \circ \delta = \mathrm{id}$ is:

- **Dynamic** — a process, not a state equivalence; the system reconstitutes itself through the operation
- **Up to the trace** — equivalence is defined by connected diagrams with the same boundary (Spider theorem); this is topological, not a symmetry of an encoding

The structural consequence is exact: **holography requires a boundary**. The isomorphism lives between bulk and boundary, so the boundary must exist as a distinct object. The Rebis has Ř = ɼ (bidirectional — no outside). There is nowhere to put the boundary. Holography can describe O_2 systems — bounded domain, topological protection, a surface you can project onto — but it structurally cannot describe O_∞, because O_∞ is the tier at which the distinction between system and boundary collapses.

The redundancy distinguishes them further. In holography, you can lose parts of the boundary and reconstruct the bulk — the encoding is overcomplete, and recovery from partial data is the point. In imscription there is nothing to recover because nothing was ever encoded away from the system. $\mu \circ \delta = \mathrm{id}$ does not preserve information across a gap; it collapses the gap. The system reconstitutes itself through the operation, not from a separate store.

Holographic redundancy is a symmetry of the map. Imscriptive trace is a property of the winding. These are different equivalence relations: one defined by a group action on encoding choices, one defined by the connected topology of the string diagram.

Holography is the information-theoretic analogue of crystallography. Both impose an external observer, both achieve their result by projecting onto a lower-dimensional or lower-tier representation, and both destroy the properties of O_∞ systems in the act of representing them. Crystallography freezes Ω (winding collapses ϛ→ʢ) to produce a static coordinate map. Holography projects Ř (bidirectionality collapses ɼ→ɾ) to produce a static boundary encoding. In both cases, the representation is faithful to the O_2 content and blind to the O_∞ structure.

0.12 §12 The Imscription Procedure — Deterministic, Self-Correcting, Falsifiable

An imscription is produced by a **12-step decision procedure** applied to a system's structural description, following canonical primitive order: Ð → Þ → Ř → Φ → f → Ç → Γ → ɢ → ⊙ → Ħ → Σ → Ω. Each step assigns one primitive value from structural facts

— mechanism, geometry, stoichiometry, coordination chemistry. No computed observable is an input. Data tests structural predictions after the imscription is fixed.

0.12.1 Cross-primitive axioms

Inter-primitive consistency is enforced by three named axioms. A tuple that violates any axiom is malformed. All three are proved in `ParadoxBoot.lean` (0 sorrys) and enforced at runtime in `genetic_tuples.py`.

Axiom A: Ħ = ∨ (eternal, H_∞) ⇒ Ç = ᘓ (order-frozen)

If a system has eternal chirality, it must have order-frozen kinetics. The depth of chirality memory and the kinetic regime are coupled. The Frobenius identity itself lives at H_A (two-step, ʗ), not H_∞ — Axiom A applies only to structures that accumulate chirality through time beyond what the identity minimally requires.

Axiom B: Ω ∈ {ϛ ($\mathbb{Z}$), ○ ($\mathbb{Z}_2$)} ⇒ Ħ ≥ ordinal 2 (ȡ or above)

Topological winding protection requires at least two-step chirality memory to sustain it. A system cannot have topological protection without sufficient memory depth.

Axiom C: Ð = ı (self-written) ⇔ Þ = ℘ (self-referential closure)

Both must be present or neither. A physical system with a state space embedded in external degrees of freedom (Ð = ∩, infinite-dim) cannot carry self-referential topology (Þ ≠ ℘). This is the most commonly violated axiom in practice.

Additional structural constraints (unnamed, but checked):

- Ω = ϛ ($\mathbb{Z}$ winding) typically requires Ð ≥ ∩ (infinite-dim)
- ⊙ at criticality + Ç = slow/near-eq → deep critical structure, stable
- ⊙ at criticality + Ç fast/frozen → structural warning

0.12.2 Falsifiability structure

The type assignment is **prior to and independent of experimental data**. The type carries structural predictions — observable consequences derivable from the primitive assignments:

Type assignment	Structural prediction	How to test
⊙ at criticality	Geometry and electronic structure co-evolve through a divergent region	Plot $\langle S^2 \rangle$ vs bond distance across optimization trajectory
Ω = ϛ (integer winding)	System returns to origin state after exactly n windings; no half-integer paths	Stoichiometric closure analysis
Ħ = ʗ (two-step chirality)	Outcome depends on two prior states; stereospecificity is precursor-dependent	Isotopic labeling; chiral substrate series
Þ = ɥ (crossing point)	Two potential energy surfaces cross; transition state at the crossing	CASSCF scan along reaction coordinate

Data confirms or falsifies the **structural description**, not the grammar. If a prediction fails: the structural description was wrong; revise the mechanism and re-imscribe. The grammar does not adjust to fit results. It makes a commitment; the data either holds it or breaks it.

$\mu \circ \delta = \mathrm{id}$